

Blue highlight = must-remember · **IMPORTANT** = asked in past papers · **BOOK p.xxx ↗** = click to open the textbook at that page · Orange box = conflict/trap between sources.

UNIT 3 · Science, Engineering & Technology in IKS

3.1 Mathematics overview

- **Gaṇita** = science of computation (arithmetic, algebra, geometry, astronomy). Features: **algorithmic approach, sūtra-style verses, integration with other sciences.**
- Phases: **Vedic** (1500–500 BCE: Śulba Sūtras) → **Classical** (500 BCE–1200 CE: Āryabhaṭa, Brahmagupta, Bhāskara I & II) → **Kerala School** (1300–1600: Mādhava, Nīlakaṇṭha).
- **Arithmetic:** Brahmagupta's rules: $a+0 = a$; $a-0 = a$; $a \times 0 = 0$; $a/0$ undefined; $(+) \times (+) = +$, $(+) \times (-) = -$, $(-) \times (-) = +$. Dhana = fortune (+), ṛṇa = debt (-). Multiplication *bhāvanā* (456×23 : $456 \times 3 = 1368$; $456 \times 20 = 9120$; sum 10488). Āryabhaṭa's square-root method (digit pairs); **Bakhshālī** square-root approximation (~200 CE). Series: $\sum n = n(n+1)/2$; $\sum n^2 = n(n+1)(2n+1)/6$; $\sum n^3 = [n(n+1)/2]^2$. AP: $a_n = a + (n-1)d$ (CIE-2 Q).

Geometry (Rekhā-gaṇita), Śulba Sūtras **IMPORTANT**

- Constructive and application-oriented (altars, temples, towns), unlike Greek axiomatic. Rope (śulba) constructions: square, rectangle, circle; shape transformations (square ↔ rectangle ↔ circle of equal area; doubling a square).
- **Baudhāyana theorem (c. 800 BCE):** "the diagonal of a rectangle produces both areas which its length and breadth produce separately" = Pythagorean theorem, centuries before Pythagoras. Triplets (3,4,5), (5,12,13).
- $\sqrt{2} \approx 1 + 1/3 + 1/(3 \times 4) - 1/(3 \times 4 \times 34) = 1.4142156$.
- Shadow geometry with gnomon (śaṅku): $h_1/s_1 = h_2/s_2$ (similar triangles).

TABLE 8.3 History of Approximations to π by Indian Mathematicians

	Value of π	Accuracy (Decimal places)	Method
Śulba-sūtras (around 800 BCE)	3.08888	1	Geometrical
Jaina texts (500 BCE)	$\sqrt{10} \approx 3.1623$	1	Geometrical
Āryabhaṭa (499 CE)	$\frac{62832}{20000} \approx 3.1416$	4	Polygon doubling ($4.2^2 = 1024$ sides)
Bhāskara II (Līlāvati) (1150 CE)	$3927/1250 \approx 3.1416$	4	Polygon doubling
Mādhava (1375 CE)	$\frac{282743338823}{9 \times 10^{11}} \approx 3.141592653592...$	11	Infinite series with end corrections
Ramanujan (1914 CE)		17 million	Modular equation

History of π (textbook Table 8.3)

IMPORTANT Value of π

- Early Vedic ≈ 3 ; Śulba ≈ 3.088 ; Jain texts $\sqrt{10} \approx 3.1623$; **Āryabhaṭa 62832/20000 = 3.1416** (4 decimals); Bhāskara II 3927/1250; **Mādhava ≈ 3.14159265359 (11 decimals)**; Ramanujan 17 million places.
- Mādhava series: $\pi/4 = 1 - 1/3 + 1/5 - 1/7 + \dots$

MQP Q32 ("approx value of π used in ancient India": $3.14 / 3 / 22/7 / \sqrt{10}$): textbook says early works used 3 and Jain works $\sqrt{10}$. If asked "ancient" with no other clue, textbook p.206 supports both **3** and **$\sqrt{10}$** ; pick whichever fits the options. 22/7 is not an Indian-text value here.

Trigonometry (Jyotpatti), Algebra (Bījagaṇita)

- India shifted from Greek **chord** to **half-chord = jyā (sine)**; *kojyā* = cosine. Āryabhaṭa: table of sine differences, 24 intervals of 3.75° ($0-90^\circ$), $R = 3438'$. Recursive sine computation; Nīlakaṇṭha finite differences; Mādhava's series for $\sin$, $\cos$, $\arctan$.
- Word history: **jyā** → **Arabic jiba/jaib** → **Latin sinus** → **"sine"**.
- Algebra**: "science of seeds" (*bīja*). Āryabhaṭa: **kuṭṭaka** (pulveriser) for $ax + c = by$. Brahmagupta: quadratic, zero and negatives (*dhana/ṛṇa*), *Brāhmasphuṭa Siddhānta* (628 CE). Bhāskara II: **chakravāla** (cyclic method) for Pell-type $Nx^2 + 1 = y^2$ ($61x^2 + 1 = y^2$: $x = 226153980$). Symbols: *yu* (add), *kṣa* (subtract), *gu* (multiply), *bhā* (divide), *mū* (root).

[BOOK p.194 Great mathematicians ↗](#)

[BOOK p.204 Geometry ↗](#)

[BOOK p.206 \$\pi\$ ↗](#)

[BOOK p.207 Trigonometry ↗](#)

[BOOK p.209 Algebra ↗](#)

[BOOK p.214 Magic squares ↗](#)

3.2 Astronomy & the Indian calendar **IMPORTANT**

Time units

Year	Basis	Length
Sāvana	Sunrise to sunrise	366 sāvana days
Saura (solar)	Sun's one zodiac round; 12 <i>rāśis</i> (≈ 30.4 days each)	≈ 365.2564 days
Cāndra (lunar)	Moon; 12 <i>māsas</i> of ≈ 29.5 days	≈ 354.367 days

- Solar–lunar gap ≈ 11 days/yr → **adhika māsa (intercalary month) about every 32–33 months (≈ 2.5 yrs)**. A 5-year small **yuga** aligns the two. **Mahāyuga** = 4 yugas (*Kṛta*, *Tretā*, *Dvāpara*, *Kali*) = **4,320,000 years**.
- Lunar month: Śukla pakṣa (waxing) + Kṛṣṇa pakṣa (waning); between two Amāvāsyā (new moon) or Pūrṇimā. Months named after the nakṣatra of the full moon (*Caitra*, *Vaiśākha*...).
- Tithi**: Moon–Sun angular distance grows by **12°** ; 30 tithis ($15 + 15$); Amāvāsyā 0° , Pūrṇimā 180° ; duration varies roughly 19–26 h (your notes print "26 h 47 min to 10 h 59 min").

IMPORTANT **Pañchāṅga: five limbs** (SEE Feb 2026 6a; MQP Q25: *Rāśi is not a limb*)

Element	Meaning	Detail
Tithi	Lunar day	12° Sun–Moon separation; fixes festivals/fasts.
Vāra	Weekday	7 days: Ravi, Soma, Maṅgala, Budha, Guru, Śukra, Śani (Sun, Moon, Mars, Mercury, Jupiter, Venus, Saturn).
Nakṣatra	Lunar mansion	27 divisions, 13°20' each (360/27); Moon stays ≈1 day in each.
Yoga	Sun + Moon longitudes	Sum ÷ 13°20' → 27 yogas ; auspicious or not.
Karaṇa	Half a tithi (6°)	11 karaṇas: 4 fixed + 7 repeating.

Uses: muhūrta (auspicious times), festivals, agriculture, navigation. Solar (Sauramāna: Tamil Nadu, Kerala) vs lunar (Cāndramāna: most of India).

As science + culture (MQP 6a): computed from Sūrya Siddhānta/Āryabhaṭīya with predictions of eclipses, equinoxes, solstices (scientific document); guides rituals, farming, social life (cultural guide).

Āryabhaṭīya (499 CE) & Siddhāntic tradition **IMPORTANT**

- **Siddhānta** = established doctrine/scientific treatise. Āryabhaṭīya: **121 verses, 4 sections**: **Ġitikāpāda** (time cycles, revolutions), **Gaṇitapāda** (maths: roots, areas, sine table), **Kālakriyāpāda** (time, calendar), **Golapāda** (sphere, eclipses, diurnal motion).
- Ideas: **Earth is spherical and rotates on its axis** (day and night; boat analogy: stationary stars appear to move backwards); eclipses are due to shadows (not demons); Moon shines by reflected light; sine function; planetary mean and true positions with **manda** and **śighra** corrections. Influence on later Indian, Arab and European astronomers.
- Siddhāntic scholars: Varāhamihira (Pañcasiddhāntikā: Pitāmaha, Vasiṣṭha, Romaka, Paulīśa, Sūrya), Bhāskara I, Brahmagupta, Lalla, Bhāskara II. Mean of 5 siddhāntas combined Indian, Greek, Babylonian ideas.

Yantras & Jantar Mantar

Yantra	Use
Śaṅku (gnomon)	Shadow length; noon, direction, latitude
Gola (armillary sphere)	Celestial coordinates; motion of stars
Nāḍivalaya	Time since sunrise, lagna (60 ghaṭikā, 12 signs)
Cakra	Sun's altitude; planetary longitude/latitude
Turiya	Quadrant; altitude and zenith distance
Ghaṭī (water clock)	Time intervals (CIE-2: the yantra for time)
Phalaka, Dhī	Altitude/zenith plank; plumb-line inclination

Jantar Mantar: Sawai Jai Singh II (1686–1743): Delhi, Jaipur, Ujjain, Varanasi, Mathura.

Instruments: Samrāt Yantra, Rāma Yantra, Jaya Prakāśa. Jyotiṣa = astronomy + astrology, a Vedāṅga; source text Vedāṅga Jyotiṣa (SEE MCQ 1.29).

3.3 Metals & metalworking **IMPORTANT**

- **Mining/extraction:** sites Zawar, Rajpura-Dariba, Khetri, Chitradurga, Singhbhum. Steps: prospecting → mining (open, adit, shaft; Zawar ≈ 500 ft) → fire-setting + chiselling → ore dressing → roasting → smelting (charcoal + flux) → refining.
- **Zinc:** India produced metallic zinc first on industrial scale, by **downward distillation at Zawar (9th–13th c. CE)**, ≈ 1000 °C; exported; used for brass.
- **Copper & alloys:** Copper (Khetri, Singhbhum, Chitradurga); **bronze = Cu + Sn**, **brass = Cu + Zn** (calamine); Copper was the first metal used (CIE-2).
- **Gold** by panning + mercury amalgamation; **silver** from galena by roasting and cupellation; **mercury** from cinnabar (Rasa-sindūra) by distillation; Ayurvedic bhasmas.
- **Apparatus (yantras):** Mūṣā (crucible), Pātana (distillation), Dhelkī (mercury), **Vāluka (sand bath: uniform heating)**, Dolā (suspension), Dhūpa (fumigation). Furnaces up to 1400 °C.

IMPORTANT Wootz steel (CIE-2, MQP 6b)

- High-carbon (**1–1.5% C**) crucible steel, South India (Tamil Nadu, Karnataka, Telangana), from ≈ 700 BCE (textbook: wootz from 4th c. BCE; CIE-2: 300 BCE–1700 CE). Name from Tamil/Kannada *ukku*.
- **Process:** wrought-iron pieces + carbon material (wood/leaves) sealed in clay crucibles, ≈ 1400 °C for ~6 h; carbon diffuses in; slow cooling forms carbides → **watered/damask pattern**. (European cementation took 6–20 days.)
- **Properties:** hard yet tough, razor-sharp, flexible, patterned.
- **Global value:** exported to Middle East/Persia/Europe; basis of legendary **Damascus swords**. Faraday studied it; precursor of alloy steels.
- **Decline (19th c.):** colonial policies/taxes, cheap European industrial steel, loss of traditional knowledge (≈ 150 years).

IMPORTANT Delhi Iron Pillar (SEE Feb 2026 6b)

- 5th c. CE; **7.2 m high, >6 tonnes, 99.7% pure wrought iron**; corrosion-free for 1,600 years.
- Why: low phosphorus + high purity; forms passive protective layer of **iron hydrogen phosphate**. Shows forge-welding, temperature and composition control.
- Other: Dhar pillar (12th c., twice as big), Konark iron beams (29), Puri beams, Thanjavur cannon, Suśruta's >100 steel surgical tools.

Iron type (Rasaratna-samuccaya)	Property	Use
Kānta-loha	Soft, magnetic, ductile	Tools, utensils
Tikṣṇa-loha (steel)	Hard, sharp, temperable	Swords, weapons
Muṣṭa-loha (cast iron)	Brittle, low melting	Casting

- **Steel making:** carburisation of wrought iron (crucible) and decarburisation of cast iron (→ mild steel). Iron extraction known from 1200 BCE; ayurvedic extraction from biotite using kaji, triphala, cow urine/milk, heated to $\approx 1400^\circ\text{C}$.

Lost-wax casting (*madhucchiṣṭa-vidhāna*, "cire perdue") **IMPORTANT**

1. Wax model (beeswax + oils/resin) with gates and vents → 2. Coat with clay (+ rice husk, sand, dung) layers, dry → 3. **Dewax:** heat, wax melts out, mould hardens → 4. Pour molten metal (bronze, brass, **pañcaloha** = Au, Ag, Cu, Sn, Zn) → 5. Break the mould; finish (chisel, polish, patina).

Each casting is unique (mould destroyed). Examples: Harappan "**Dancing Girl**" (Mohenjo-Daro, 2500 BCE), **Chola bronzes** (9th–12th c.; Naṭarāja), Bastar, Odisha. Somesvara's *Mānasollāsa* (12th c.) describes it. Still done at Swamimalai. Modern use: jewellery, turbine blades.

[BOOK p.253 Wootz ↗](#)
[BOOK p.255 Mining ↗](#)
[BOOK p.259 Zinc ↗](#)
[BOOK p.265 Iron & steel ↗](#)
[BOOK p.271 Steel manufacture ↗](#)
[BOOK p.272 Lost wax ↗](#)
[BOOK p.273 Apparatus ↗](#)

3.4 Dyes & painting

- Sources: **plants** (indigo = blue, madder = red, turmeric = yellow, saffron), **minerals** (ochres, malachite, azurite), **insects** (lac, cochineal), metals (orpiment). **Mordants** (alum, iron salts, lime, tannins) fix colour; Mohenjo-Daro dyed cotton fragment proves mordant knowledge.
- Methods: direct dyeing; mordant dyeing (madder + alum = red, + iron = purple); **resist dyeing** (Bandhani, Ikat); overdyeing (turmeric + indigo = green). **Indigo** (*Indigofera tinctoria*): leaves fermented to indigotin; India main supplier till 19th c. (MQP Q33: leaf).
- Painting: murals (Ajanta, Ellora, Sittanavasal, Bagh) on lime plaster; miniatures; fresco; folk (Madhubani, Warli, Pattachitra). Binders: gum arabic, resins, egg white, casein. Words *calico*, *chintz*.

3.5 The art of making perfumes (*Gandhaśāstra* / *Gandhayukti*)

IMPORTANT

- **Sources:** flowers (jasmine, rose, champaka), woods (sandal, agarwood), herbs (vetiver, camphor), resins (benzoin), spices, animal (musk, ambergris).
- **Methods:** **deg-bhapka steam distillation** (copper still *deg* + receiver *bhapka*; still used for **attar at Kannauj**); maceration/infusion in oil or fat; distillation of woods/resins; blending with **fixatives** (musk, sandal oil); ageing.
- **Products:** sugandhi taila, attar, dhūpa/agarbatti, gandha-chūrṇa, rose/vetiver/saffron water.
- **Texts:** Varāhamihira's *Br̥hat Saṃhitā* chapter 77 "**Gandha-yukti**" (37 verses; uses permutations: 16 ingredients choose 4 = 1,820 combos × 96 = 174,720 perfumes); Gandhasāra; Charaka/Suśruta. 8-step process *karmāṣṭakam*: śodhana, vāsana, virecana, bhāvanā, pāka, bodhana, dhūpana, vedhana.
- **Significance:** ritual, medicine, cosmetics, status/hospitality; global trade; "attar" ← Sanskrit *gandha* via Persian *itr*.



Chand Baori stepwell (textbook Fig 11.3)



Grand Anicut / Kallanai, world's oldest dam still in use

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[BOOK p.295 Perfumes ↗](#)